Branch: MCA(AI & ML)	**Semester:3rd**
Student Name: Harshita Sharma	**UID:25MCI10337**
Subject Name: Technical training-2	**Subject Code: 25CAP-707**
Section/Group: MAM-2(B)	**Date of Performance: 18 Sep, 2026**

Experiment No-3:

1. Aim:

To extend the responsive food delivery interface by developing backend services and integrating with a database.

Tasks: • Design and implement RESTful APIs for Restaurants, Menu, and Cart operations

• Create backend using Node.js and Express

• Connect application with database (MySQL)

• Implement CRUD operations for: • Restaurants • Menu items

2. Project Structure :

```
cruspoo-backend/
├── server.js          Express app entry point
├── db.js              MySQL connection pool
├── schema.sql         CREATE DATABASE / TABLE / INSERT script
├── package.json
├── .env.example
└── routes/
    ├── restaurants.js  CRUD for restaurants
    ├── menu.js         CRUD for menu items
    └── orders.js       Cart checkout + order history

cruspoo-frontend/
└── index.html         original file — UI unchanged, fetch() added
```

3. Code for experiment/Practical:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Cruspo | Food Delivery System</title>
  <!-- Bootstrap 5 & FontAwesome CDN -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.2/css/all.min.css" rel="stylesheet">
  <style>
    :root { --primary-color: #ff5e14; }
    .navbar-brand { font-weight: 800; color: var(--primary-color) !important; }
    .btn-primary-custom { background-color: var(--primary-color); color: #fff; border: none; }
    .food-card { border: none; border-radius: 12px; overflow: hidden; transition: transform 0.2s; }
```

```
      .food-card:hover { transform: translateY(-4px); }
      .food-card-img { height: 180px; object-fit: cover; }
      .cart-overlay { display: none; position: fixed; top: 0; left: 0; width: 100%; height: 100%; background: rgba(0,0,0,0.5); z-index: 1045; }
      .cart-overlay.active { display: block; }
    </style>
</head>
<body>
    <!-- 1. NAVIGATION BAR -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark sticky-top shadow-sm">
        <div class="container">
            <a class="navbar-brand d-flex align-items-center gap-2" href="#"><i class="fa-solid fa-utensils"></i> Cruspo</a>
            <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarContent">
                <span class="navbar-toggler-icon"></span>
            </button>
            <div class="collapse navbar-collapse" id="navbarContent">
                <ul class="navbar-nav me-auto">
                    <li class="nav-item"><a class="nav-link active" href="#offers">Offers</a></li>
                    <li class="nav-item"><a class="nav-link" href="#restaurants">Restaurants</a></li>
                    <li class="nav-item"><a class="nav-link" href="#menu">Menu</a></li>
                </ul>
                <div class="d-flex align-items-center gap-3">
                    <input class="form-control" type="search" id="searchInput" placeholder="Search menu..." onkeyup="filterMenu()">
                    <button class="btn btn-outline-light position-relative" onclick="toggleCart(true)">
                        <i class="fa-solid fa-cart-shopping fs-5"></i>
                        <span id="cartCountBadge" class="position-absolute top-0 start-100 translate-middle badge rounded-pill bg-danger">0</span>
                    </button>
                </div>
            </div>
        </div>
    </nav>
    <!-- 2. HERO CAROUSEL -->
    <section id="offers" class="py-4 container">
        <div id="offersCarousel" class="carousel slide rounded-4 overflow-hidden shadow" data-bs-ride="carousel">
            <div class="carousel-inner">
                <div class="carousel-item active">
                    <img src="https://images.unsplash.com/photo-1585032226651-759b368d7246?auto=format&fit=crop&w=1200&q=80" class="d-block w-100" style="height: 350px; object-fit: cover;">
                    <div class="carousel-caption bg-dark bg-opacity-50 p-3 rounded">
                        <h2>Special Offers - Up to 30% OFF</h2>
                    </div>
                </div>
            </div>
        </div>
    </section>
    <!-- 3. MENU & FILTER SECTION -->
    <section id="menu" class="py-4 container">
        <h3 class="fw-bold mb-3">Explore Menu</h3>
        <!-- Filter Controls Grid -->
```

```
        <div class="row g-3 mb-4 bg-white p-3 rounded shadow-sm align-items-center">
          <div class="col-md-3">
            <select id="dietFilter" class="form-select" onchange="filterMenu()">
              <option value="all">All Dietary Types</option>
              <option value="veg">Vegetarian</option>
              <option value="non-veg">Non-Vegetarian</option>
            </select>
          </div>
          <div class="col-md-3">
            <select id="cuisineFilter" class="form-select" onchange="filterMenu()">
              <option value="all">All Categories</option>
              <option value="Chinese">Chinese</option>
              <option value="South Indian">South Indian</option>
              <option value="Street Food">Street Food</option>
            </select>
          </div>
          <div class="col-md-4">
            <label class="form-label mb-0 fw-bold">Max Price: ₹<span id="priceValue">500</span></label>
            <input type="range" class="form-range" id="priceFilter" min="40" max="500" step="10" value="500"
oninput="document.getElementById('priceValue').innerText=this.value; filterMenu();">
          </div>
          <div class="col-md-2">
            <button class="btn btn-outline-secondary w-100" onclick="resetFilters()">Reset</button>
          </div>
        </div>
        <!-- Dynamic Food Cards Container -->
        <div id="menuContainer" class="row g-4"></div>
      </section>
      <!-- 4. SHOPPING CART MODAL & BACKDROP -->
      <div id="cartOverlay" class="cart-overlay" onclick="toggleCart(false)"></div>
      <div class="modal" id="cartModal" tabindex="-1" style="z-index: 1050;">
        <div class="modal-dialog modal-dialog-centered modal-lg">
          <div class="modal-content">
            <div class="modal-header bg-dark text-white">
              <h5 class="modal-title fw-bold"><i class="fa-solid fa-cart-shopping me-2"></i> Your Cart</h5>
              <button type="button" class="btn-close btn-close-white" onclick="toggleCart(false)"></button>
            </div>
            <div class="modal-body">
              <div id="cartEmptyMessage" class="text-center py-4"><p class="text-muted">Your cart is empty.</p></div>
              <div id="cartItemsContainer" class="table-responsive d-none">
                <table class="table align-middle">
                  <thead><tr><th>Item</th><th>Price</th><th>Qty</th><th>Subtotal</th><th>Action</th></tr></thead>
                  <tbody id="cartTableBody"></tbody>
                </table>
              </div>
              <div id="cartSummary" class="border-top pt-3 d-none">
                <div class="d-flex justify-content-between fs-5 fw-bold p-2 bg-light rounded">
                  <span>Total Amount:</span>
                  <span id="cartTotal" class="text-danger">₹0</span>
                </div>
```

```
            </div>
          </div>
          <div class="modal-footer">
            <button class="btn btn-outline-secondary" onclick="toggleCart(false)">Close</button>
            <button id="checkoutBtn" class="btn btn-success" onclick="checkout()" disabled>Checkout</button>
          </div>
        </div>
      </div>
    </div>
    <!-- Bootstrap 5 JS -->
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/bootstrap.bundle.min.js"></script>
</body>
</html>
```

4. Rest API Refernce :

5.1 Restaurants

Method	Endpoint	Description
GET	/api/restaurants	List all restaurants
GET	/api/ restaurants/:id	Get a single restaurant
POST	/api/restaurants	Create a new restaurant
PUT	/api/ restaurants/:id	Update an existing restaurant
DELETE	/api/ restaurants/:id	Delete a restaurant

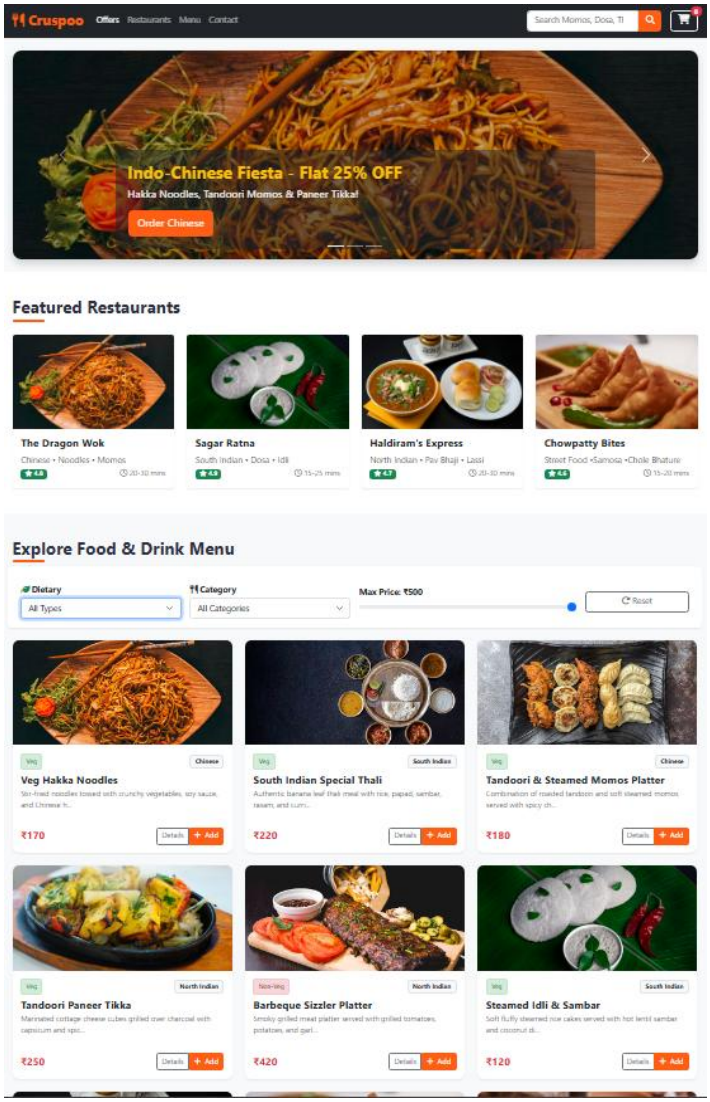
5.2 Menu

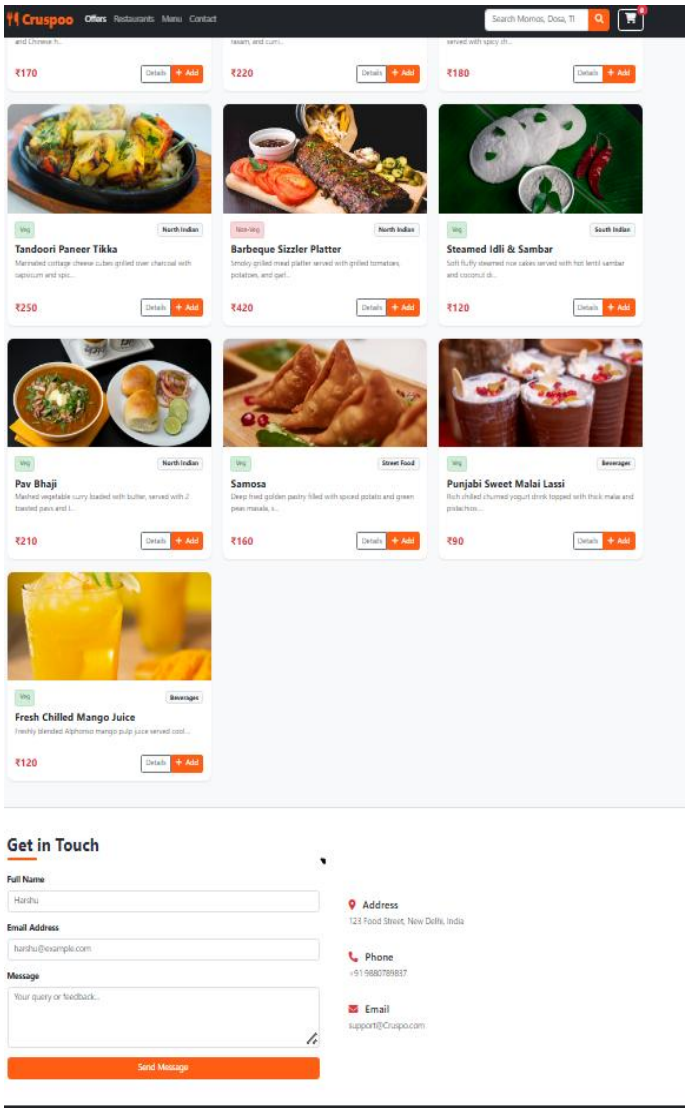
Method	Endpoint	Description
GET	/api/menu	List all menu items (filters: restaurant_id, category, cuisine)
GET	/api/menu/:id	Get a single menu item
POST	/api/menu	Create a new menu item
PUT	/api/menu/:id	Update an existing menu item
DELETE	/api/menu/:id	Delete a menu item

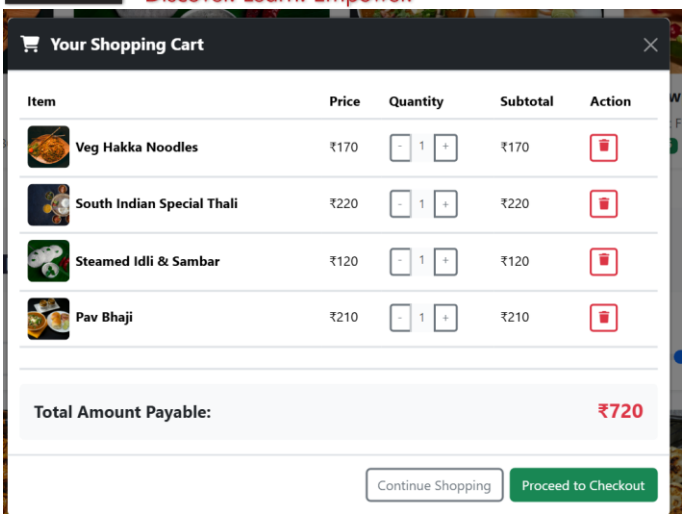
5.3 Cart / Orders

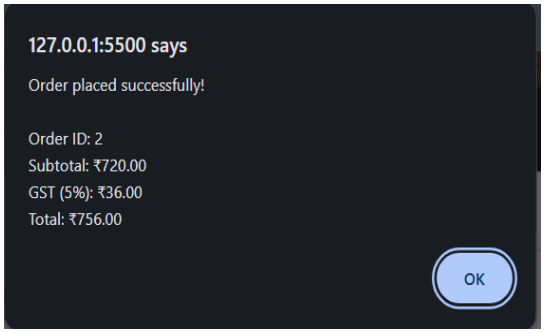
Method	Endpoint	Description
POST	/api/orders	Place an order from cart items: { items: [{menu_id, quantity}] }
GET	/api/orders	List all past orders
GET	/api/orders/:id	Get one order with its line items

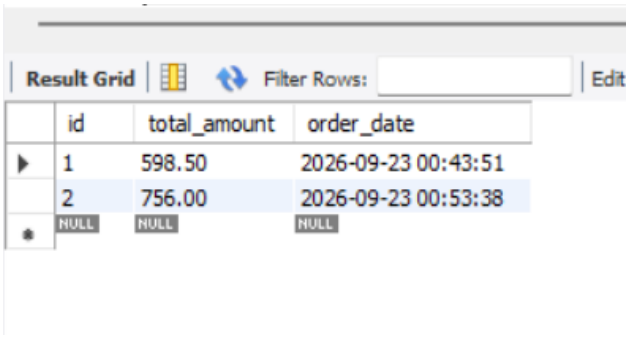
5. Result/Output/Writing Summary:











6. Learning outcomes (What I have learnt):

- Backend Architecture: Learned how to design and build robust RESTful APIs using Node.js and Express to handle asynchronous client-server communication.
- Database Integration: Gained the practical ability to connect web applications with a relational database Management system (MySQL) to persistently store and retrieve application data.
- CRUD Operations: Mastered the implementation of Create, Read, Update, and Delete operations across core modules including restaurants, menu catalogs, and user shopping carts.
- Order & Transaction Management: Understood how to structure and process customer checkout workflows, handling multi-variable records such as delivery addresses, total amounts, and payment methods.